

Generalization of Methods for Voltage-Sag Source Detection Using Vector-Space Approach

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Abstract—This paper discusses methods for voltage-sag source detection, which are based either on energy, current, or impedance criteria. It is shown that methods known from the literature do not work well, particularly in cases of asymmetrical voltage sags. Therefore, generalized methods for voltage-sag source detection are proposed, using a vector-space approach. All the discussed methods, both the already known and the proposed ones, were tested by applying extensive simulations, as well as laboratory and field tests. The correctness of the proposed generalized methods was verified in this way. The obtained results show that all the proposed methods are highly effective in all cases of voltage sags, whereas the overall effectiveness of those methods known from literature is mainly unacceptable.

Index Terms—Power quality, power systems, simulation, source detection, testing, voltage sag.

I. INTRODUCTION

VOLTAGE SAGS are the most frequent among the wide range of power-quality disturbances, since they can be provoked by different events throughout the network, such as faults, motor starting, transformer energizing (TR-E), and heavy-load switching [1]. Despite their relatively short duration—usually less than 1 s [2]—voltage sags may be detrimental to several industrial loads. The impact of disturbances caused by voltage sags on production losses have already been reported [3]–[5], as well as the severe influence of voltage sags on the behavior of induction machines and three-phase transformers [6], [7]. The detection and measurement of voltage sags is, therefore, essential for possible mitigation [8]–[10], as well as for further analysis [11].

Reliable information about a voltage-sag source is indispensable in order to identify the responsible party for production losses or interruptions in the power supply. Even though a methodology for pinpointing the exact locations of voltage sags does not exist yet, several methods for voltage-sag source detection have already been reported [12]–[16]. The method proposed in [12] is based on the assumption that the energy

flow at the monitoring point (MP) increases during downstream events and decreases during upstream events. The methods proposed in [13] and [14] are both based on the assumption that currents measured at the MP change (increase or decrease) during the voltage sag. The slope of a current–voltage trajectory is investigated in [13], whereas, in [14], the time-behavior of an active-current component is observed. In [15] and [16], two impedance-based methods are proposed, where the real part of the estimated impedance [15] or the impedance angle [16] is used to determine the direction of the voltage-sag source.

The testing of all the discussed methods for voltage-sag source detection known from the literature shows that in the cases of asymmetrical voltage sags, these methods are rather ineffective [17]. Furthermore, all the discussed methods, except the energy-based one [12], require computation of voltage and current phasors for the fundamental-frequency component. Because voltage sags are transient disturbance events, all phasor-based methods might produce questionable results due to the inherent averaging in the harmonic analysis of the input signals.

A theory of vector-space approach [18], [19] is, therefore, applied in order to overcome these difficulties. Thus, several new methods for voltage-sag source detection are introduced. All the proposed methods are based on instantaneous voltages and currents; however, only instantaneous Clarke's $\alpha\beta$ components are taken into account. All the discussed methods for voltage-sag source detection were tested by applying extensive simulations, laboratory tests, and field tests. The results for different types of power-system faults, heavy motor starting and loading, TR-E, and distributed generation were analyzed, in order to evaluate all the discussed methods for voltage-sag source detection. The correctness and superior performances of the proposed generalized methods were verified in this way.

II. VECTOR-SPACE APPROACH—MATHEMATICAL BACKGROUND

Voltage sags are transient disturbance events where voltages and currents are generally asymmetrical. Moreover, they might contain a considerable amount of higher sub- and interharmonics, as well as aperiodic components. Conventional phasor-based methods are, therefore, inappropriate for the analysis of voltage sags. Proper detection of voltage-sag sources can be performed using a vector-space approach [18]. Therefore, several instantaneous quantities are introduced, such as instantaneous voltage and current vectors, instantaneous active power ($\alpha\beta$ and zero sequence), and instantaneous equivalent

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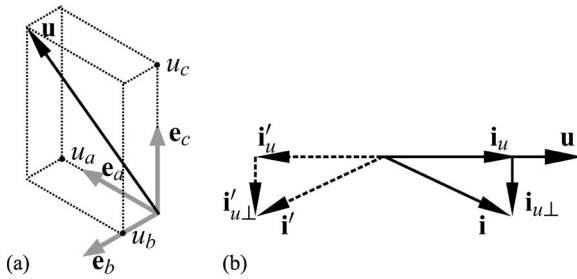


Fig. 1. Graphic presentation of the (a) voltage vector and (b) orthogonal projections of the current vector—considering (b-dashed case) orientation of the active-current vector.

conductivity, which can be used for the generalization of phasor-based methods for voltage-sag source detection.

A. Instantaneous Voltage and Current Vectors, Active-Power Definition, and Orthogonal Decomposition of Currents

The voltages and currents of a three-phase system can be handled using abstract vector space [19]. A vector space with an inner product $\langle \cdot, \cdot \rangle$ is called a Euclidean space. Let us introduce the orthonormal basis $\mathcal{B}_{abc}$ into the Euclidean space by (1) where each of the basis vectors $\mathbf{e}_a$, $\mathbf{e}_b$, and $\mathbf{e}_c$ represents an individual phase of a three-phase system

$$\mathcal{B}_{abc} = \{\mathbf{e}_a, \mathbf{e}_b, \mathbf{e}_c\}, \quad \langle \mathbf{e}_j, \mathbf{e}_k \rangle = \begin{cases} 1, & \text{if } j = k \\ 0, & \text{if } j \neq k. \end{cases} \quad (1)$$

The voltage and current vectors are defined by (2), where $u_a(t)$, $u_b(t)$, $u_c(t)$ and $i_a(t)$, $i_b(t)$, $i_c(t)$ denote line voltages and currents, respectively. The voltage vector is shown in Fig. 1(a)

$$\begin{aligned} \mathbf{u}(t) &:= \mathbf{e}_a u_a(t) + \mathbf{e}_b u_b(t) + \mathbf{e}_c u_c(t) \\ \mathbf{i}(t) &:= \mathbf{e}_a i_a(t) + \mathbf{e}_b i_b(t) + \mathbf{e}_c i_c(t). \end{aligned} \quad (2)$$

Let us define the inner product $\langle \cdot, \cdot \rangle$ as a mapping of two vectors into a scalar quantity, where several properties must be fulfilled [18]. The Euclidean norm is defined by the inner product. The Euclidean norm is a scalar whose value is equal to the vector's length. The norms of the voltage and the current vectors are defined by

$$\begin{aligned} \|\mathbf{u}(t)\| &:= \sqrt{\langle \mathbf{u}(t), \mathbf{u}(t) \rangle} = \sqrt{u_a^2(t) + u_b^2(t) + u_c^2(t)} \\ \|\mathbf{i}(t)\| &:= \sqrt{\langle \mathbf{i}(t), \mathbf{i}(t) \rangle} = \sqrt{i_a^2(t) + i_b^2(t) + i_c^2(t)}. \end{aligned} \quad (3)$$

Furthermore, the instantaneous active power of a three-phase system is defined by the inner product of the voltage and the current vector

$$p(t) := \langle \mathbf{u}(t), \mathbf{i}(t) \rangle = u_a(t)i_a(t) + u_b(t)i_b(t) + u_c(t)i_c(t). \quad (4)$$

The current vector can be expressed as the sum of two vectors, as shown in Fig. 1(b). They are the projection of the current vector on the voltage vector (5), and the projection of the current vector orthogonal (perpendicular) to the voltage

vector (6) [20]. According to (4) and (5), the instantaneous equivalent conductivity of a three-phase system is defined by

$$\mathbf{i}_u(t) := \frac{\langle \mathbf{u}(t), \mathbf{i}(t) \rangle}{\|\mathbf{u}(t)\|^2} \mathbf{u} = G_e(t) \mathbf{u}(t) \quad (5)$$

$$\mathbf{i}_{u\perp}(t) := \mathbf{i}(t) - \mathbf{i}_u(t) \quad (6)$$

$$G_e(t) := \frac{p(t)}{\|\mathbf{u}(t)\|^2}. \quad (7)$$

Vectors $\mathbf{i}_u(t)$ and $\mathbf{i}_{u\perp}(t)$ are orthogonal, therefore, their inner product is equal to zero $\langle \mathbf{i}_u(t), \mathbf{i}_{u\perp}(t) \rangle = 0$. Thus, the values of inner products $\langle \mathbf{u}(t), \mathbf{i}(t) \rangle$, and $\langle \mathbf{u}(t), \mathbf{i}_u(t) \rangle$ are equal. According to (8), the instantaneous active power can be obtained with the vector $\mathbf{i}(t)$ or with the projection $\mathbf{i}_u(t)$. On this basis, the vector $\mathbf{i}_u(t)$ can be called “the instantaneous active-current vector” [20]. The norm of the active-current vector is expressed by

$$\langle \mathbf{u}(t), \mathbf{i}(t) \rangle = \langle \mathbf{u}(t), \mathbf{i}_u(t) \rangle = p(t) \quad (8)$$

$$\|\mathbf{i}_u(t)\| = \frac{|p(t)|}{\|\mathbf{u}(t)\|}. \quad (9)$$

Let us consider the case where the orientations of the active-current vector and the voltage vector are opposite [dashed case in Fig. 1(b)]. In this case, (9) does not preserve the vector's orientation, since its value is always positive. However, according to (5), the equivalent conductivity (7) is negative for the discussed case ($G_e(t) < 0$). The orientation of the discussed norm is thus considered by introducing the factor $S = \text{sign}(G_e(t))$, as expressed in

$$S \|\mathbf{i}_u(t)\| = \text{sign}(G_e(t)) \|\mathbf{i}_u(t)\| = \frac{p(t)}{\|\mathbf{u}(t)\|}. \quad (10)$$

B. Clarke's Components

The voltage vector $\mathbf{u}(t)$ and the current vector $\mathbf{i}(t)$ may be expressed using any basis in the Euclidean space. Let us introduce a new orthonormal basis $\mathcal{B}_{\alpha\beta 0}$, thus, the voltage and the current vectors are expressed as

$$\begin{aligned} \mathbf{u}(t) &= \mathbf{e}_\alpha u_\alpha(t) + \mathbf{e}_\beta u_\beta(t) + \mathbf{e}_0 u_0(t) \\ \mathbf{i}(t) &= \mathbf{e}_\alpha i_\alpha(t) + \mathbf{e}_\beta i_\beta(t) + \mathbf{e}_0 i_0(t). \end{aligned} \quad (11)$$

Voltages and currents $u_\alpha(t)$, $u_\beta(t)$, $u_0(t)$ and $i_\alpha(t)$, $i_\beta(t)$, $i_0(t)$ are defined by (12), where $\mathbf{C}$ denotes the orthogonal Clarke's transformation matrix (13)

$$\begin{bmatrix} u_\alpha(t) \\ u_\beta(t) \\ u_0(t) \end{bmatrix} = \mathbf{C} \begin{bmatrix} u_a(t) \\ u_b(t) \\ u_c(t) \end{bmatrix} \quad \begin{bmatrix} i_\alpha(t) \\ i_\beta(t) \\ i_0(t) \end{bmatrix} = \mathbf{C} \begin{bmatrix} i_a(t) \\ i_b(t) \\ i_c(t) \end{bmatrix} \quad (12)$$

$$\mathbf{C} = \sqrt{\frac{2}{3}} \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \quad \mathbf{C}^{-1} = \mathbf{C}^T. \quad (13)$$

Note that in the three-phase three-wire system, the voltage and current vectors always lie in the plane defined by the basis

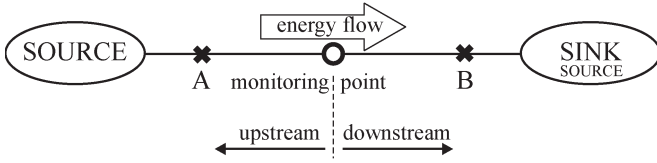


Fig. 2. Upstream event (A) and downstream event (B).

vectors $\mathbf{e}_\alpha$ and $\mathbf{e}_\beta$, except in cases, such as phase-to-ground faults, where the sum of the line voltages and the sum of the line currents are not zero. Let us, therefore, also define vectors $\mathbf{u}_{\alpha\beta}(t)$ and $\mathbf{i}_{\alpha\beta}(t)$ by

$$\begin{aligned}\mathbf{u}_{\alpha\beta}(t) &= \mathbf{e}_\alpha u_\alpha(t) + \mathbf{e}_\beta u_\beta(t) \\ \mathbf{i}_{\alpha\beta}(t) &= \mathbf{e}_\alpha i_\alpha(t) + \mathbf{e}_\beta i_\beta(t).\end{aligned}\quad (14)$$

Because the transformation matrix is orthogonal ($\mathbf{C}^{-1} = \mathbf{C}^T$), the norms of the original and the transformed vectors are equal and can also be expressed by

$$\begin{aligned}\|\mathbf{u}(t)\| &= \sqrt{u_\alpha^2(t) + u_\beta^2(t) + u_0^2(t)} \\ \|\mathbf{i}(t)\| &= \sqrt{i_\alpha^2(t) + i_\beta^2(t) + i_0^2(t)}.\end{aligned}\quad (15)$$

The norms of the voltage and current vectors $\mathbf{u}_{\alpha\beta}(t)$ and $\mathbf{i}_{\alpha\beta}(t)$ are defined by

$$\begin{aligned}\|\mathbf{u}_{\alpha\beta}(t)\| &= \sqrt{u_\alpha^2(t) + u_\beta^2(t)} \\ \|\mathbf{i}_{\alpha\beta}(t)\| &= \sqrt{i_\alpha^2(t) + i_\beta^2(t)}.\end{aligned}\quad (16)$$

The transformation (12) is power invariant. Thus, the instantaneous active power can also be expressed by (17), where $p_0(t)$ is the instantaneous zero-sequence power whereas $p_{\alpha\beta}(t)$ is the instantaneous $\alpha\beta$ -sequence power [21]

$$\begin{aligned}p(t) &= (u_\alpha(t)i_\alpha(t) + u_\beta(t)i_\beta(t)) + u_0(t)i_0(t) \\ &= p_{\alpha\beta}(t) + p_0(t).\end{aligned}\quad (17)$$

The projection of the current vector $\mathbf{i}_{\alpha\beta}(t)$ on the voltage vector $\mathbf{u}_{\alpha\beta}(t)$ is defined by (18). The equivalent $\alpha\beta$ -sequence conductivity and the norm of the active current vector are introduced by (19) and (20), respectively, where $S_{\alpha\beta} = \text{sign}(G_{e,\alpha\beta}(t))$

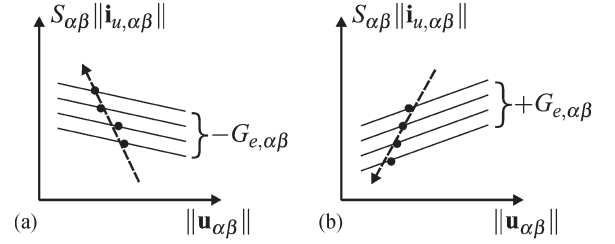
$$\mathbf{i}_{u,\alpha\beta}(t) := G_{e,\alpha\beta}(t)\mathbf{u}_{\alpha\beta}(t) \quad (18)$$

$$G_{e,\alpha\beta}(t) := \frac{p_{\alpha\beta}(t)}{\|\mathbf{u}_{\alpha\beta}(t)\|^2} \quad (19)$$

$$S_{\alpha\beta} \|\mathbf{i}_{u,\alpha\beta}(t)\| = \frac{p_{\alpha\beta}(t)}{\|\mathbf{u}_{\alpha\beta}(t)\|}. \quad (20)$$

III. GENERALIZATION OF PHASOR-BASED METHODS FOR VOLTAGE-SAG SOURCE DETECTION

Let us consider the MP, as shown in Fig. 2. Voltage sags might originate either from point A or from point B. With regard to the presag energy-flow direction in the steady state,

Fig. 3. Characteristic $(\|\mathbf{u}_{\alpha\beta}(t)\|, S_{\alpha\beta} \|\mathbf{i}_{u,\alpha\beta}(t)\|)$: (a) downstream event and (b) upstream event.

upstream and downstream events are defined in points A and B, respectively. A power-quality monitor or another recording device is placed at the MP. Based on the recorded voltages $u_k(t)$ and currents $i_k(t)$, where $k \in \{a, b, c\}$ (a, b , and c denote individual phases), it is possible to determine on which side of the recording device the voltage sag originated. Several methods for voltage-sag source detection have already been reported [12]–[16]; however, most of them require calculations of phasors for the fundamental-frequency component of voltages and currents. Therefore, the definitions introduced in the previous section are used to generalize phasor-based methods for voltage-sag source detection [13]–[16]. In this way, generalized methods are developed, which are all based on instantaneous voltage and current vectors, whereas only Clarke's $\alpha\beta$ components are taken into account.

A. Generalization of the Voltage–Current Method

Voltage sags are due to short-duration increases in currents elsewhere in the network. Thus, currents measured at the MP increase during downstream events and decrease during upstream events. The phasor-based voltage–current method is proposed in [13] based on this assumption, where the points of $(\hat{I}, |\hat{U} \cos \phi|)$ are approximated using a linear function for each phase individually, in order to investigate its slope (21). $\hat{U}$ and $\hat{I}$ are voltage and current phasor lengths, respectively, whereas ϕ is a phase angle

$$\text{slope}(\hat{I}, |\hat{U} \cos \phi|) \begin{cases} > 0 \Rightarrow \text{upstream} \\ < 0 \Rightarrow \text{downstream.} \end{cases} \quad (21)$$

This method can be generalized by applying the norms of the voltage and current vectors. However, the norm of the active-current vector (20) is used, since only this current-vector component contributes to the energy transmission [20], where the energy-flow direction is considered by the factor $S_{\alpha\beta}$. Each point of the characteristic $(\|\mathbf{u}_{\alpha\beta}(t)\|, S_{\alpha\beta} \|\mathbf{i}_{u,\alpha\beta}(t)\|)$ is therefore at a certain line with slope of $-G_{e,\alpha\beta}$ for downstream events [Fig. 3(a)], or at a certain line with slope of $+G_{e,\alpha\beta}$ [Fig. 3(b)] for upstream events. Thus, the generalized voltage–current method can be introduced by (22), where the points of $(\|\mathbf{u}_{\alpha\beta}(t)\|, S_{\alpha\beta} \|\mathbf{i}_{u,\alpha\beta}(t)\|)$ are approximated using the linear function

$$\text{slope}(\|\mathbf{u}_{\alpha\beta}(t)\|, S_{\alpha\beta} \|\mathbf{i}_{u,\alpha\beta}(t)\|) \begin{cases} > 0 \Rightarrow \text{upstream} \\ < 0 \Rightarrow \text{downstream.} \end{cases} \quad (22)$$

B. Generalization of the Active-Current-Based Method

The method proposed in [14] is based on the same mathematical background as the method of (21) [13]. Within the discussed method, the time response of an active-current component ($\hat{I} \cos \phi$) is calculated for a few cycles prior to and during the voltage sag. The sign of its first peak at the beginning of the voltage sag is used as the criterion (23) for each phase individually. $\hat{I}$ is a current phasor length, whereas ϕ is a phase angle

$$\text{first peak}(\hat{I} \cos \phi) \begin{cases} < 0 \Rightarrow \text{upstream} \\ > 0 \Rightarrow \text{downstream.} \end{cases} \quad (23)$$

In order to generalize this method, the norm of the active-current vector (20) can be used as the criterion. In this way, the generalized active-current-based method is introduced by (24), where the time response of $(S_{\alpha\beta} \|\mathbf{i}_{u,\alpha\beta}(t)\|)$ is calculated for a few cycles prior to and during the voltage sag, i.e.,

$$\text{first peak} (S_{\alpha\beta} \|\mathbf{i}_{u,\alpha\beta}(t)\|) \begin{cases} < 0 \Rightarrow \text{upstream} \\ > 0 \Rightarrow \text{downstream.} \end{cases} \quad (24)$$

C. Generalization of Impedance-Based Methods

The concept of incremental impedance is proposed in [15], which is negative for downstream events and positive for upstream events. However, only the real part of the estimated impedance is used as the criterion (25). Incremental impedance is obtained from the incremental voltage and current phasors for the positive-sequence component, which are defined as $\Delta \underline{U} = (\underline{U}_{\text{sag}} - \underline{U}_{\text{presag}})$ and $\Delta \underline{I} = (\underline{I}_{\text{sag}} - \underline{I}_{\text{presag}})$, respectively, whereas $\Delta \underline{Z} = (\Delta \underline{U} / \Delta \underline{I})$. In order to improve the estimated impedance, adaptive multiple cycles of data are used by applying the least-squares method

$$\text{Real}(\Delta \underline{Z}) = \Delta R \begin{cases} > 0 \Rightarrow \text{upstream} \\ < 0 \Rightarrow \text{downstream.} \end{cases} \quad (25)$$

Another impedance-based method is proposed in [16], which is based on the assumption that the estimated impedance during the voltage sag changes both in magnitude $|\underline{Z}|$ and in angle $\angle \underline{Z}$. Thus, criterion (26) is introduced, where the results obtained prior to and during the voltage sag are compared, i.e.,

$$\begin{aligned} &\text{if } |\underline{Z}_{\text{sag}}| < |\underline{Z}_{\text{presag}}| \text{ and } \angle \underline{Z}_{\text{sag}} > 0 \Rightarrow \text{downstream} \\ &\text{else} \Rightarrow \text{upstream.} \end{aligned} \quad (26)$$

Note that criteria (25) and (26) are very similar. Therefore, only the resistance-sign-based method (25) [15] is discussed in this paper.

Let us introduce the concept of equivalent conductivity (19) instead, which is negative for downstream events [negative slope in Fig. 3(a)], whereas it is positive for upstream events [positive slope in Fig. 3(b)]. According to Fig. 3, the incremental conductivity is defined by the ratio (27), where incremental norms of the voltage and active-current vectors are defined as $\Delta \|\mathbf{u}_{\alpha\beta}(t)\| = (\|\mathbf{u}_{\alpha\beta}(t)\|_{\text{sag}} - \|\mathbf{u}_{\alpha\beta}(t)\|_{\text{presag}})$ and $\Delta (S_{\alpha\beta} \|\mathbf{i}_{u,\alpha\beta}(t)\|) =$

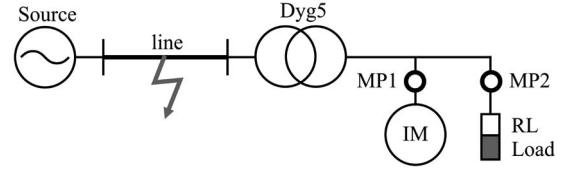


Fig. 4. Radial network example during an upstream fault.

$((S_{\alpha\beta} \|\mathbf{i}_{u,\alpha\beta}(t)\|)_{\text{sag}} - (S_{\alpha\beta} \|\mathbf{i}_{u,\alpha\beta}(t)\|)_{\text{presag}})$, respectively, i.e.,

$$\Delta G_{e,\alpha\beta}(t) := \frac{\Delta (S_{\alpha\beta} \|\mathbf{i}_{u,\alpha\beta}(t)\|)}{\Delta \|\mathbf{u}_{\alpha\beta}(t)\|}. \quad (27)$$

The obtained expression (27) might lead to numerical problems due to possible division by zero and, therefore, cannot be used as a criterion. However, only the sign of the ratio (27) is needed for the voltage-sag source-detection criterion, which is positive only when the signs of the numerator and denominator are equal, otherwise it is negative. Thus, the generalized resistance (conductivity)-sign-based method is proposed by

$$\frac{\text{sign}(\text{first peak}(\Delta \|\mathbf{u}_{\alpha\beta}(t)\|))}{\text{sign}(\text{first peak}(\Delta (S_{\alpha\beta} \|\mathbf{i}_{u,\alpha\beta}(t)\|)))} \begin{cases} > 0 \Rightarrow \text{upstream} \\ < 0 \Rightarrow \text{downstream.} \end{cases} \quad (28)$$

D. Energy-Based Method

Voltage sags are provoked by different events which can all be treated as energy sinks. Thus, it can be assumed that the energy flow at the MP increases during downstream events and decreases during upstream events. In order to detect the voltage-sag source, the disturbance energy $\Delta w(t)$ is used as the criterion (29) [12], where the disturbance power is defined by the difference $\Delta p(t) := (p(t)_{\text{sag}} - p(t)_{\text{presag}})$, i.e.,

$$\Delta w(t) := \int_0^t \Delta p(\tau) d\tau \begin{cases} < 0 \Rightarrow \text{upstream} \\ > 0 \Rightarrow \text{downstream.} \end{cases} \quad (29)$$

The new energy-based method (30) is introduced by considering only the $\alpha\beta$ -sequence instantaneous power, where the disturbance power is defined by the difference $\Delta p_{\alpha\beta}(t) := (p_{\alpha\beta}(t)_{\text{sag}} - p_{\alpha\beta}(t)_{\text{presag}})$

$$\Delta w_{\alpha\beta}(t) := \int_0^t \Delta p_{\alpha\beta}(\tau) d\tau \begin{cases} < 0 \Rightarrow \text{upstream} \\ > 0 \Rightarrow \text{downstream.} \end{cases} \quad (30)$$

IV. RESULTS

A. Advantages of Generalized Methods

In order to present the advantages of the proposed generalized methods for voltage-sag source detection, a radial network is considered during a fault on the supply side, as shown in Fig. 4. On the customer side, a balanced RL load and an induction motor (IM) are considered, whereas line voltages and currents are captured at two MPs (MP1 and MP2). Thus, the upstream source of voltage sags is observed at MP1 and MP2.

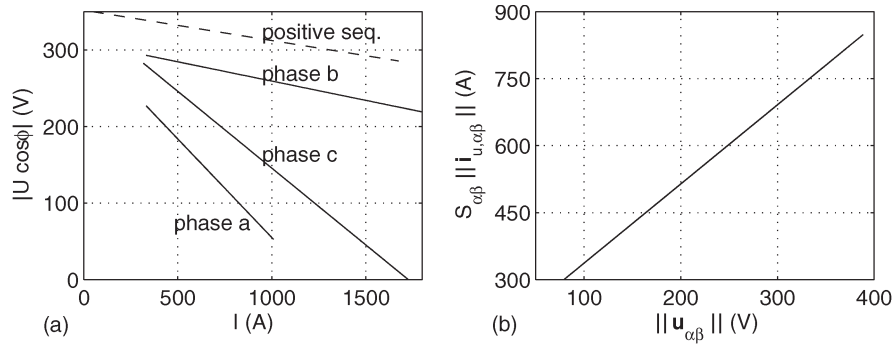


Fig. 5. Trajectories obtained by the voltage-current method for the upstream phase *a*-to-phase *b* fault (Fig. 4-MP1): (a) phasor-based method and (b) generalized method.

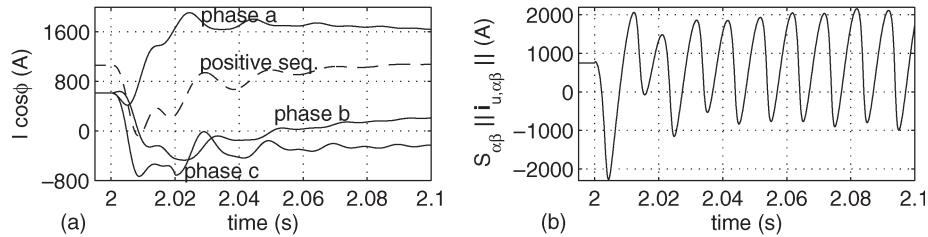


Fig. 6. Time responses obtained by the active-current-based method for the upstream phase *a*-to-phase *b* fault (Fig. 4-MP1): (a) phasor-based method and (b) generalized method.

1) *Asymmetrical Voltage Sags*: The criteria within the phasor-based methods (21) [13] and (23) [14] are checked for each phase individually. In the cases of asymmetrical voltage sags, individual phases show different behavior. Consequently, the obtained results may be inconclusive. Note that in such cases, the usage of positive-sequence component for voltages and currents may be helpful; yet, in [13] and [14], this is not suggested. However, the proposed generalized methods (22) and (24) are expected to give more reliable results, since they are based on instantaneous values whereas the results are always conclusive. Let us, therefore, consider a radial network shown in Fig. 4 during the upstream phase *a*-to-phase *b* fault. According to the *ABC* classification [22], a type *C* voltage sag is provoked (only in phases *a* and *b*). Note that the transformer Dyg5 is placed between the voltage-sag source and the MP (MP1 is considered). Thus, a type *D* voltage sag is observed at MP1, i.e., a small sag in phases *a* and *b* and a large sag in phase *c*.

The calculated trajectories obtained by the phasor-based voltage-current method (21) and the generalized voltage-current method (22) are shown in Fig. 5(a) and (b), respectively. It can be seen from the results shown in Fig. 5(a) that the obtained slopes are negative for all individual phases, which incorrectly indicates a downstream event. Moreover, the result is incorrect even when using only positive-sequence component for voltages and currents [dashed line in Fig. 5(a)]. From the result obtained by the generalized method [Fig. 5(b)], it is obvious that the slope is positive, which is a correct result, i.e., upstream for this case.

Results obtained by both active-current-based methods, the phasor-based (23) and the generalized one (24), are shown in Fig. 6(a) and (b), respectively. Time responses in Fig. 6(a)

show different signs of the first peak for individual phases (positive peak in phase *a* and negative peaks in phases *b* and *c*). The so-obtained results cannot be interpreted in a unique way and are, therefore, inconclusive. By applying the positive-sequence quantities, the result is conclusive and correct, since the negative first peak in the obtained time response indicates upstream event [dashed line in Fig. 6(a)]. The time response obtained by the generalized method is also correct, since the first peak is negative [Fig. 6(b)], which indicates upstream event.

2) *Signal Distortions*: Most of the methods for voltage-sag source detection, (21) [13], (23) [14], (25) [15], and (26) [16], require fundamental-harmonic components of sampled voltages and currents, which are extracted using discrete orthogonal series expansion (Fourier or Walsh). Note that algorithms of this type are particularly appropriate for studying steady-state and periodical signals. Voltage sags are, on the contrary, transient events. Thus, usage of the discussed algorithms may be inappropriate. Performance of the discussed methods was, therefore, tested under severe supply-voltage distortion. Let us again consider the network shown in Fig. 4 during the upstream phase *a*-to-phase *b* fault, where the voltage sag is captured at MP1. Furthermore, higher, sub- and interharmonics are assumed for the “Source” as follows: $u(t) = (0.1u_{28\text{Hz}}(t) + 1u_{50\text{Hz}}(t) + 0.1u_{100\text{Hz}}(t) + 0.2u_{150\text{Hz}}(t) + 0.1u_{117\text{Hz}}(t))$ (p.u.).

The results obtained by both voltage-current methods (21) and (22) are similar to those obtained for the nominal supply voltage [Fig. 5(a) and (b)]. The phasor-based method gives incorrect results, whereas only the generalized method correctly indicates upstream event.

Time responses obtained by both active-current-based methods (23) and (24) are shown in Fig. 7(a) and (b), respectively.

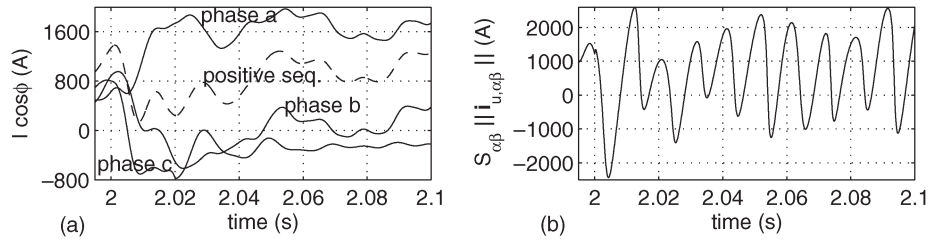


Fig. 7. Time responses obtained by the active-current-based method for the upstream phase *a*-to-phase *b* fault (Fig. 4-MP1)—perturbed supply voltage: (a) phasor-based method and (b) generalized method.

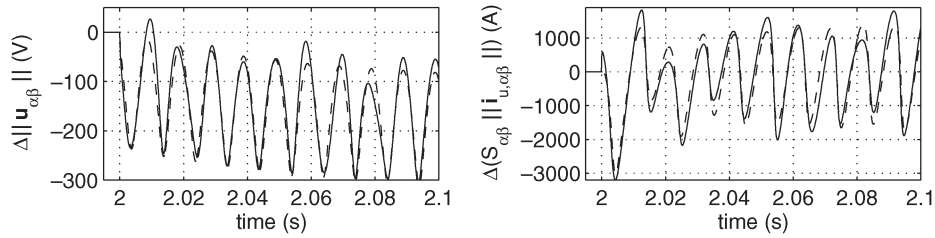


Fig. 8. Time responses obtained by the generalized resistance-sign-based method for the upstream phase *a*-to-phase *b* fault (Fig. 4-MP1): (solid line) perturbed supply voltage and (dashed line) nominal supply voltage.

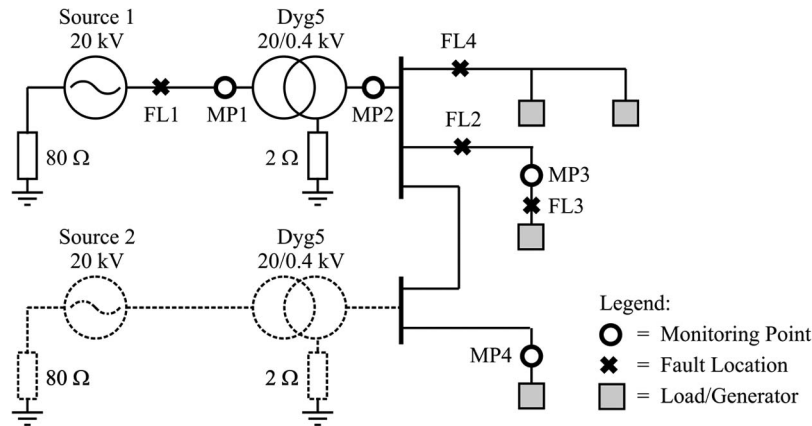


Fig. 9. Testing-network for numerical simulations of voltage sags: (solid line) radial network and (solid and dashed line) nonradial network.

The time responses obtained by the phasor-based method show different signs of the first peak for individual phases and, therefore, cannot be interpreted in a unique way. However, when positive-sequence quantities are applied, no particular peak is expressed in the obtained time response [dashed line in Fig. 7(a)], which is due to the averaging of the input signals. On the contrary, the time response obtained by the generalized current-based method is conclusive and correct (upstream event), since the first peak is negative, as shown in Fig. 7(b).

The influence of signal distortions within the resistance-sign-based methods (25) and (28) is discussed next. When using the phasor-based method for the case with nominal supply voltage, the positive incremental resistance is determined ($+0.024 \Omega$), which correctly indicates an upstream event, whereas the negative value of incremental resistance is determined for the case with perturbed supply voltage (-0.034Ω), which is incorrect for this case. However, the generalized resistance-sign-based method shows correct results (upstream) in both cases, since the signs of first peaks in the incremental voltage-vector norm and the incremental active-vector norm are equal, as shown in Fig. 8.

3) Upstream-Fault Events in Radial Networks: In a radial network and for upstream faults, there might be no change in the seen impedance, neither in magnitude nor in angle. In such cases, both impedance-based methods (25) [15] and (26) [16] can produce unreliable results.

Let us consider the network shown in Fig. 4, where the voltage sag is captured at MP2. The incremental resistance obtained for this case is indeed positive ($+1.6 \Omega$). However, the value of the voltage-current ratio does not change in time. Thus, the resistance-sign-based method (25) cannot be used in this case. The generalized method (28), on the other hand, gives conclusive and correct result (upstream event) as well as all the other phasor-based and generalized methods.

B. Evaluation of Phasor-Based and Generalized Methods

Evaluation of the discussed methods for voltage-sag source detection was performed by applying extensive numerical simulations of voltage sags. A radial and nonradial testing-network, as shown in Fig. 9, was selected for numerical simulations of voltage sags. An extensive number of tests were performed

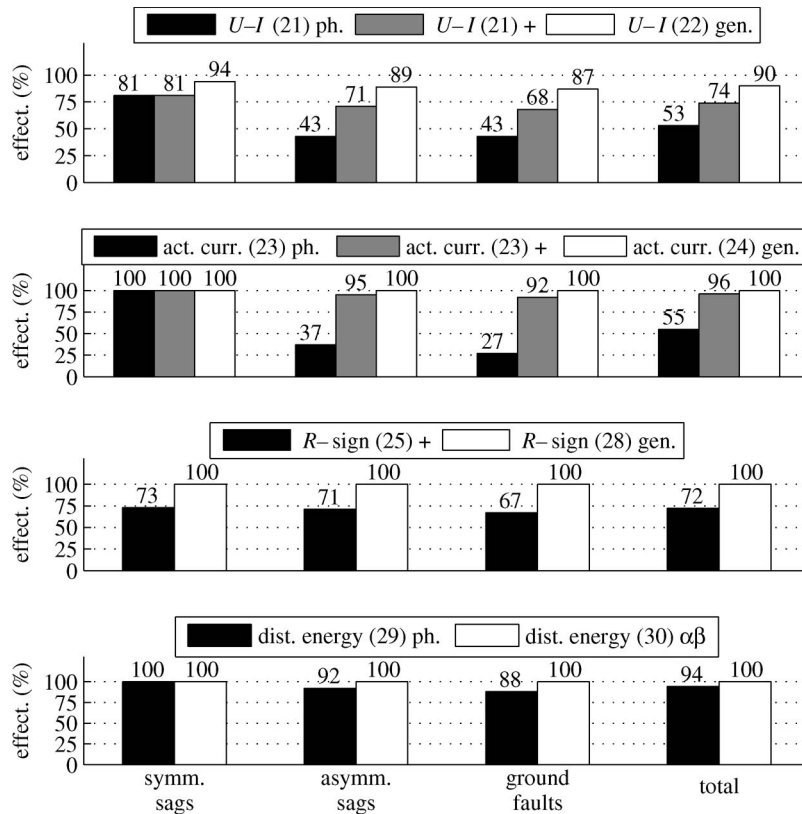


Fig. 10. Effectiveness of the discussed methods for voltage-sag source detection. Legend: (·) equation number, “ph.” phase values, “+” positive sequence, “ $\alpha\beta$ ” Clarke’s, “gen.” generalized method.

for different combinations of loads and for different events using MATLAB/Simulink. Different types of loads, such as balanced RL load, IM, synchronous motor, as well as constant-power loads were used. Small generators, such as induction generator (IG) and synchronous generator, were also used in order to simulate the distributed generation. Four types of faults were applied in four different locations (FL1–FL4): phase-to-ground fault (P–G), phase-to-phase-to-ground fault (P–P–G), phase-to-phase fault (P–P), and three-phase fault (3–P). In this way, voltage sags were generated with a sufficient magnitude ($> 15\%$) and duration (100 ms). Furthermore, voltage sags due to heavy motor starting and loading were also simulated. Voltages and currents were captured at four MPs (MP1–MP4) using a sampling frequency of 10 kHz.

Altogether, 174 different examples of voltage sags were analyzed, whereas effectiveness was determined for all the discussed methods. Symmetrical voltage sags due to 3–P faults, motor loading, and motor starting were separately examined (48 examples), as well as asymmetrical voltage sags due to P–G faults, P–P–G faults, and P–P faults (126 examples). Special attention was paid also to voltage sags due to P–G faults (84 examples). The obtained results are shown in Fig. 10. The results shown in Fig. 10 show that all generalized methods (denoted by “gen.”), except the voltage–current ($U-I$) method, gave us correct direction of the voltage sag in all 174 examples. On the contrary, the phasor-based methods, particularly the voltage–current and active-current methods, show very low effectiveness when the phase quantities (denoted by “ph.”) were used. However, when the positive-sequence quantities (denoted

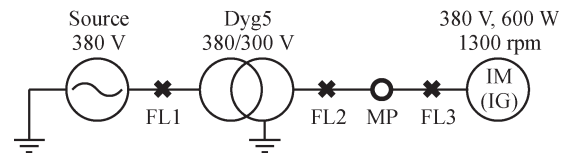


Fig. 11. Laboratory setup for generation of voltage sags.

by “+”) were used, the effectiveness of the active-current method was surprisingly high. The phasor-based resistance-sign-method shows quite low effectiveness, which is mostly due to the upstream faults where the impedance was not changed during the sag. Note that all the discussed methods show almost the same effectiveness for voltage sags captured in the radial network and those captured in the nonradial network.

C. Laboratory and Field-Testing Results

An experimental laboratory setup for the generation of voltage sags was built, as shown in Fig. 11, in order to confirm the discussed methods for voltage-sag source detection. An IM was used as an active load, whereas an IG was also used to simulate a small power plant. Different types of faults were performed at three different locations (FL1–FL3): P–G fault, P–P–G fault, P–P fault, and 3–P fault. In order to generate faults, time-controlled relays were used, whereas the fault duration was set to 100 ms. A sufficient magnitude of the voltage sag ($> 15\%$) was obtained by adjusting the fault resistance with

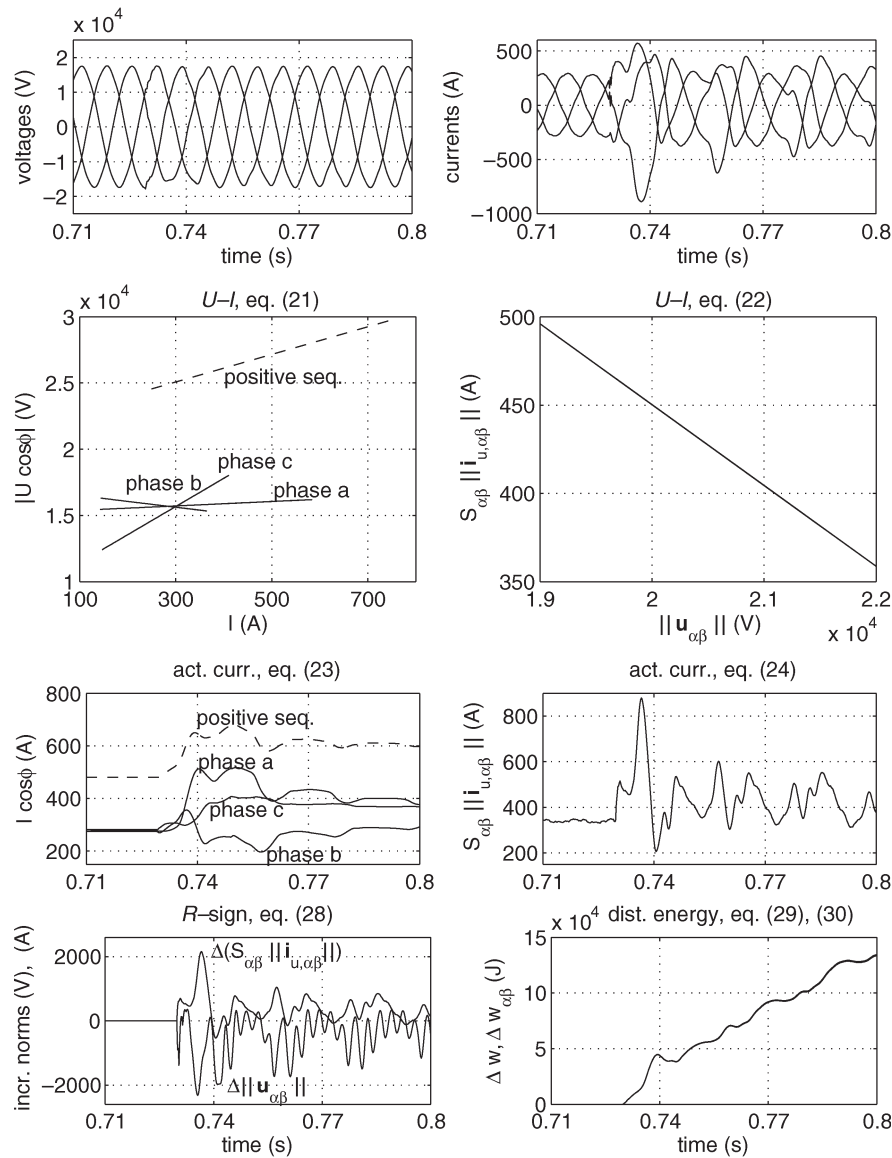


Fig. 12. Field-testing results for the downstream TR-E.

additional resistors. Voltage sags due to IM loading were also generated. Voltages and currents were captured at the MP by a digital control system dSPACE DS1103 PPC, where a sampling frequency of 10 kHz was used.

Furthermore, an extensive number of field tests was also applied. Voltages and currents were captured in a Slovenian power system with a sampling frequency of 5 kHz. An example of field testing results is shown in Fig. 12, where a P-G fault was generated within the 20-kV network. This event led to the protection-relay trip and, then, to the successful auto reclosure. Subsequently, a power transformer was energized. A downstream voltage sag was thus provoked at the 20-kV bus, where voltages and currents were captured. The obtained results (Fig. 12) show high distortions in captured currents, which is typical for the TR-E. Even though the magnitude of the discussed voltage sag was small ($\sim 5\%$), all generalized methods correctly indicated downstream event. The results obtained by the phasor-based voltage-current method and active-current method were inconclusive, since they show different behav-

ior for different phases. However, when the positive-sequence quantities were used, the result obtained by the active-current method was correct (positive peak), whereas the result obtained by the voltage-current method was not (positive slope). The phasor-based resistance-sign-method correctly indicated downstream event, since the obtained resistance was negative (-8.85Ω).

All field-testing and laboratory-testing results obtained by the discussed methods are given in Tables I and II, respectively, and are in complete accordance with the simulation-based result shown in Fig. 10. Laboratory-testing results 1–11 are given for the case when the IM was used as an active load, whereas test 12 refers to the voltage sag generated by IM loading. Furthermore, results 13–23 are given for the case when a small power plant with IG was tested. Field-testing results were captured at different voltage levels (0.4, 20, 110, 220, and 400 kV). Tests 1–14 are due to different types of power-system faults, whereas tests 15 and 16 are due to TR-E. Note that in Fig. 12, results of field test 15 are shown.

TABLE I
LABORATORY-TESTING RESULTS

test no.	event source	fault loca.	$U-I$		act. curr.		R -sign		dist. ener.	
			ph.+ gen.	ph.+ gen.	ph.+ gen.	ph.+ gen.	+ gen.	+ gen.	ph. $\alpha\beta$	ph. $\alpha\beta$
			(21)	(22)	(23)	(24)	(25)	(28)	(29)	(30)
1	↑P-G	FL1	○↑	↑	○×	↑	↑	↑	↑	↑
2	↑P-P-G	FL1	○↑	↑	○×	↑	↑	↑	↑	↑
3	↑P-P	FL1	○↑	↑	○×	↑	↑	↑	↑	↑
4	↑P-G	FL2	↑↑	↑	↑↑	↑	↑	↑	↑	↑
5	↑P-P-G	FL2	↑↑	↑	↑↑	↑	↑	↑	↑	↑
6	↑P-P	FL2	↑↑	↑	↑↑	↑	↑	↑	↑	↑
7	↑3-P	FL2	↑↑	↑	↑↑	↑	↑	↑	↑	↑
8	↓P-G	FL3	×	×	↓	↓	×	↓	↓	↓
9	↓P-P-G	FL3	×	×	↓	↓	×	↓	↓	↓
10	↓P-P	FL3	×	×	↓	↓	×	↓	↓	↓
11	↓3-P	FL3	×	×	↓	↓	×	↓	↓	↓
12	↓IM-L	—	×	×	↓	↓	↓	↓	↓	↓
13	↓P-G	FL1	○×	↓	○×	↓	↓	↓	↓	↓
14	↓P-P-G	FL1	○×	↓	↓×	↓	↓	↓	↓	↓
15	↓P-P	FL1	○×	↓	○×	↓	↓	↓	↓	↓
16	↓P-G	FL2	×	×	○×	↓	↓	↓	↓	↓
17	↓P-P-G	FL2	○×	↓	○×	↓	↓	↓	↓	↓
18	↓P-P	FL2	○×	↓	○×	↓	↓	↓	↓	↓
19	↓3-P	FL2	↓×	↓	↓×	↓	↓	↓	↓	↓
20	↑P-G	FL3	↑↑	↑	○×	↑	↑	↑	↑	↑
21	↑P-P-G	FL3	○↑	↑	○×	↑	↑	↑	↑	↑
22	↑P-P	FL3	○↑	↑	○×	↑	↑	↑	↑	↑
23	↑3-P	FL3	×	↑	↑×	↑	↑	↑	↑	↑

Legend: ↓ downstream, ↑ upstream, ○ not conclusive, × incorrect, (·) eq. no., “ph.” phase values, “+” pos. seq., “ $\alpha\beta$ ” Clarke’s, “gen.” generalized method.

TABLE II
FIELD-TESTING RESULTS

test no.	event source	MP (kV)	$U-I$		act. curr.		R -sign		dist. ener.	
			ph.+ gen.	ph.+ gen.	ph.+ gen.	ph.+ gen.	+ gen.	+ gen.	ph. $\alpha\beta$	ph. $\alpha\beta$
			(21)	(22)	(23)	(24)	(25)	(28)	(29)	(30)
1	↓P-G	20	○×	×	○↓	↓	×	↓	↓	↓
2	↓P-G	20	○×	×	○↓	↓	×	↓	↓	↓
3	↓P-G	110	○↓	↓	○↓	↓	↓	↓	↓	↓
4	↓P-G	110	○↑	↑	○↑	↑	↑	↑	↑	↑
5	↓P-G	110	○↑	↑	○↑	↑	↑	↑	↑	↑
6	↓P-G	110	○↓	↓	○↓	↓	↓	↓	↓	↓
7	↓P-G	110	○×	↑	○↑	↑	↑	↑	↑	↑
8	↑P-G	220	○×	↑	○×	↑	×	↑	↑	↑
9	↓P-G	400	○↓	↓	↓×	↓	↓	↓	↓	↓
10	↓P-P	20	○↓	↓	↓↓	↓	↓	↓	↓	↓
11	↓P-P	20	○↓	↓	○○	↓	○	↓	↓	↓
12	↑P-P	110	↑↑	↑	○↑	↑	↑	↑	↑	↑
13	↓3-P	20	↓↓	↓	↓↓	↓	↓	↓	↓	↓
14	↑3-P	110	×	↑	↑×	↑	↑	↑	↑	↑
15	↓TR-E	20	○×	↓	○↓	↓	↓	↓	↓	↓
16	↑TR-E	0.4	↑↑	↑	○○	↑	↑	↑	↑	↑

Legend: ↓ downstream, ↑ upstream, ○ not conclusive, × incorrect, (·) eq. no., “ph.” phase values, “+” pos. seq., “ $\alpha\beta$ ” Clarke’s, “gen.” generalized method.

V. CONCLUSION

This paper has presented and evaluated eight methods for voltage-sag source detection. The three already-known phasor-based methods are generalized using a vector-space approach, whereas the new energy-based method is introduced by considering only $\alpha\beta$ -sequence components. All the discussed methods, the already known and the proposed ones, were extensively tested applying numerical simulations, laboratory tests, and field tests. Based on the performed evaluation results, it can

be concluded that all the discussed methods are very effective in cases of symmetrical voltage sags. However, in cases of asymmetrical voltage sags, the proposed generalized methods are much more effective than the phasor-based methods, which was completely confirmed by both laboratory- and field-testing results.

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